

AUTHORS

Mohid Tahir, Faiyaz Faruq and Felix Chen

AFFILIATIONS

University of Toronto
MIE Undergraduate Department
MIE315 - Design for the Environment
Professor Kevin Golovin
TA Sara Mohsenitaromsari

Google Home Speaker: Wooden or Plastic Housing?

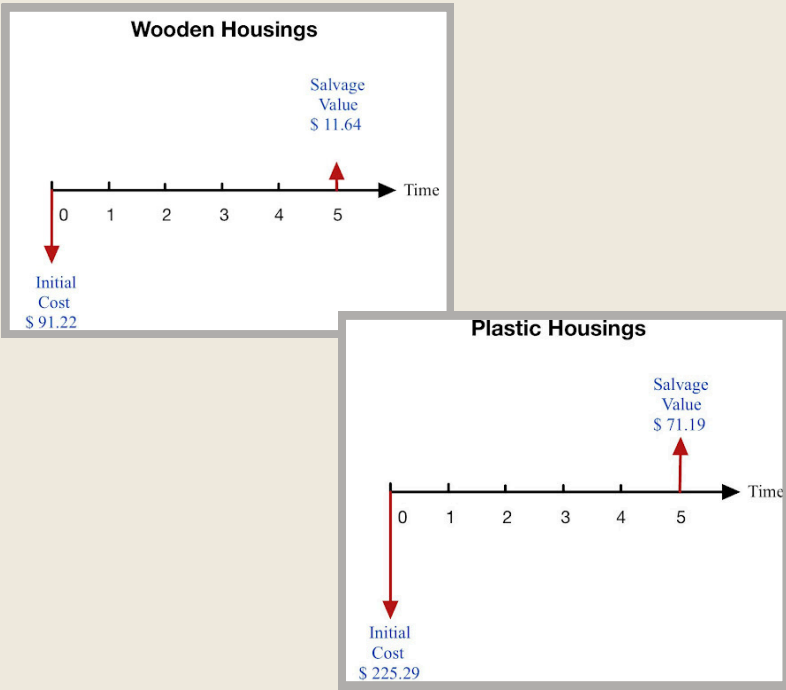


Functional Unit:
per 100 units of Google Home Speaker Housings

Evaluation of the benefits and drawbacks associated with transitioning the Google Home Speaker’s housing material from plastic to wood.

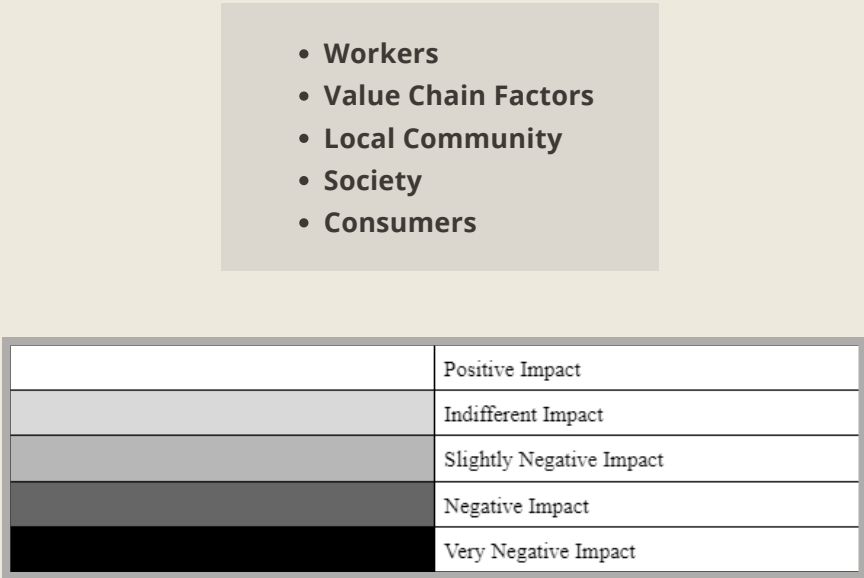
Financial Recommendation

Although neither option provides a positive net present value, the Wooden option results in less money lost, and a focus towards manufacturing.



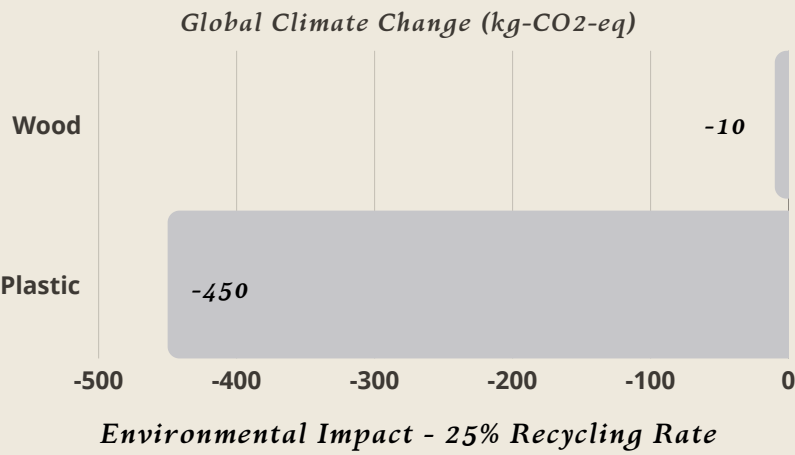
Societal Recommendation

The analysis compared plastic and wood for Google's speaker housing, evaluating societal impacts like safety and employment. Despite weaknesses in both, wood was favored for its positive societal impact.



Uncertainty Analysis

The uncertainty analysis evaluates how assumptions in the life cycle and financial analysis impact results. Variations in speaker weight could lower emissions but require structural adjustments. Recycling assumptions significantly affect environmental and health impacts, highlighting the benefits of efficient recycling for plastic. Financial uncertainties from fluctuating interest rates have minimal effect on the overall conclusions, indicating the analysis's resilience to such changes.



Functions, Objectives and Constraints

- Primary Function:** Voice-controlled interaction with the internet and smart home devices.
- Objectives:**
- Environmental Impacts.
 - Economic Implications
 - User Acceptance
 - Structural integrity
- Constraints:**
- Weight Limit: < 1.2kg
 - Cost Limit: < \$150
 - Durability: > 1 year

DfE Considerations for Design

- Low production output
- Brittle and soft pine wood, resulting in higher failures
- Energy requirements of machining with wood

Solution Approach

- Redesign the Google Home structure for wood focused manufacturing
- Utilize Engineered woods
 - Extend life
 - Allow recycled items to be better utilized
- Utilize waste from machining to power production processes
- Create Engineered woods internally

Hybrid LCA

Pre-Manufacturing

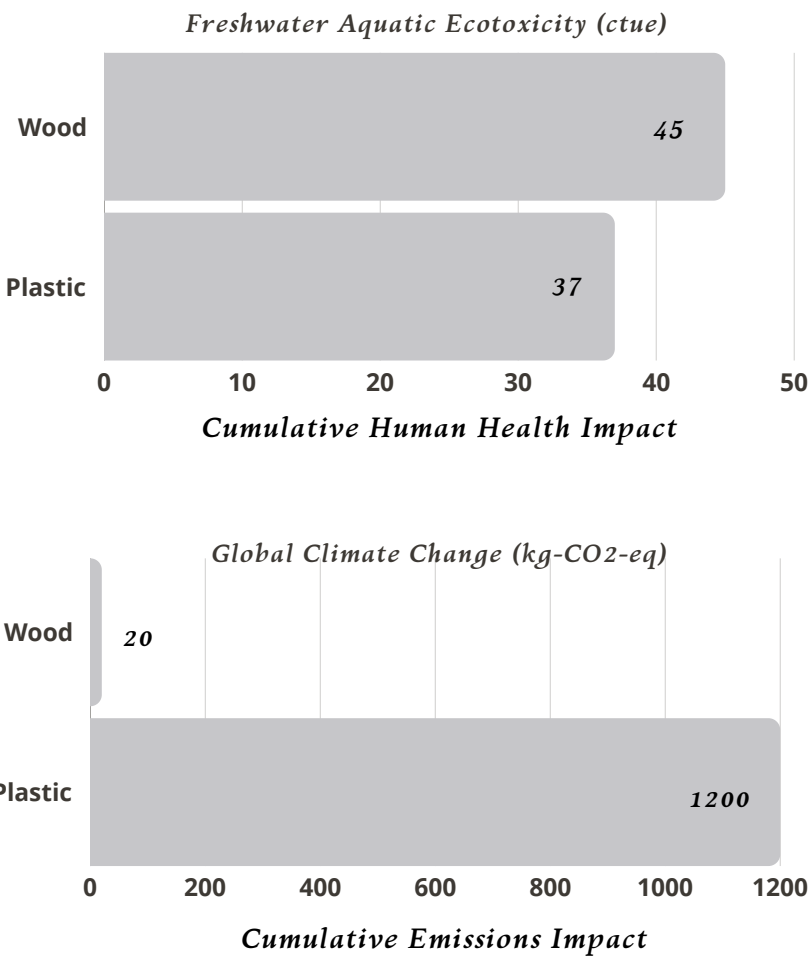
Wood impacts land and produces waste; plastic doubles climate change effects. Wood preferred despite higher ecotoxicity due to lower climate impact.

Manufacturing

Plastic's environmental and health impacts significantly exceed wood's. Wood's minimal effects make it the better choice for manufacturing.

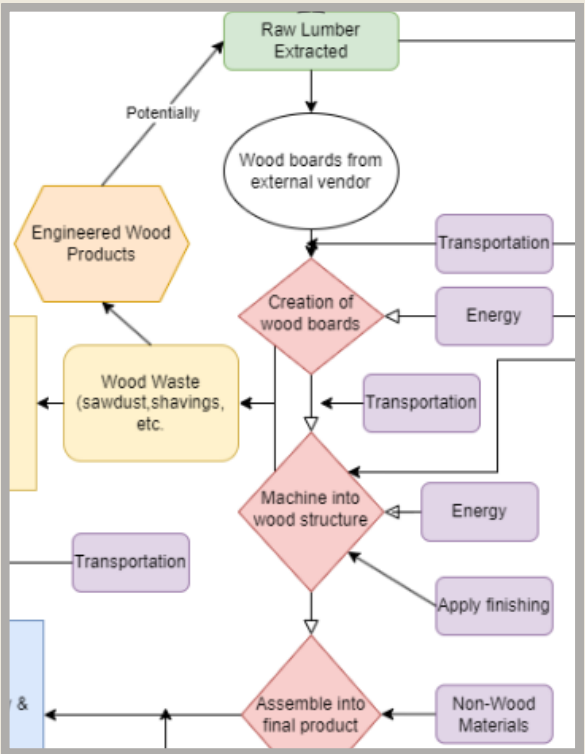
Disposal

Recycling's greater negative impact pushes wood to be favourable, despite it being compromised by varnish coating's ecotoxicity in landfills.



Final Recommendation

The financial analysis favors wood over plastic for smart speaker housings due to a less negative NPV. Life Cycle Assessment shows wood has lower impacts on climate change and smog, despite slightly higher freshwater ecotoxicity. Societal analysis highlights wood's benefits in craftsmanship, sustainability, community health, and consumer perception, outweighing plastic's advantages. Despite uncertainties, conclusions are deemed resilient, affirming wood as the preferred choice for environmental, financial, and societal reasons.



References

